



KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY
Indian Institute of Science campus, Bengaluru

Telephone: 080 – 23341652, 23348848 || Email: spp@kscst.org.in
Website: www.kscst.org.in/spp.html or https://kscst.karnataka.gov.in/en

**FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE
49th SERIES OF STUDENT PROJECT PROGRAMME**

<https://forms.gle/iXiR81MyoLWn871x7>

1.	Name of the College: Adichunchanagiri Institute of Technology, Chikkamagaluru
2.	Project Title: “Multi Class Classification of Lung Diseases on Chest X-ray Images Using Deep learning”
3.	Branch: Information Science and Engineering
4.	Theme (as per KSCST poster): Big Data, IoT, Artificial Intelligence, Data Engineering, Robotics (in agriculture, health, education, digital literacy, assistive aid)
5.	Name(s) of project guide(s): 1. Name: Mr. Rakesh S Raj Email id: rakeshraj.ait@gmail.com Contact No.: +91 9902770688 2. Name: Mr. Ravishankar R Email id: ravishankar.r.gowda@gmail.com Contact No.: +91 9980318999

6.	Name of Team Members (Strictly not more than four students in a batch): <i>(Type names in Capital Letters as provided in your college)</i> (Please paste the latest passport size photograph adjacent to your respective names) <table border="1"> <tr> <td data-bbox="233 331 938 613"> Name: BHAVYA N U USN No.: 4AI22IS008 Email id: bhavyabhavyanu08@gmail.com Mobile No: +91 8618915473 </td> <td data-bbox="938 331 1294 613">  </td> </tr> <tr> <td data-bbox="233 613 938 896"> Name: NAGAMMA V T USN No.: 4AI22IS025 Email id: nagammatimmanagoudra@gmail.com Mobile No.: +91 8073339817 </td> <td data-bbox="938 613 1294 896">  </td> </tr> <tr> <td data-bbox="233 896 938 1182"> Name: SHASHANK C L USN No.: 4AI22IS041 Email id: shashank07cl@gmail.com Mobile No.: +91 8618025986 </td> <td data-bbox="938 896 1294 1182">  </td> </tr> <tr> <td data-bbox="233 1182 938 1476"> Name: VASUDHA B M USN No.: 4AI22IS061 Email id: vasudham282@gmail.com Mobile No.: +91 9740164456 </td> <td data-bbox="938 1182 1294 1476">  </td> </tr> </table>	Name: BHAVYA N U USN No.: 4AI22IS008 Email id: bhavyabhavyanu08@gmail.com Mobile No: +91 8618915473		Name: NAGAMMA V T USN No.: 4AI22IS025 Email id: nagammatimmanagoudra@gmail.com Mobile No.: +91 8073339817		Name: SHASHANK C L USN No.: 4AI22IS041 Email id: shashank07cl@gmail.com Mobile No.: +91 8618025986		Name: VASUDHA B M USN No.: 4AI22IS061 Email id: vasudham282@gmail.com Mobile No.: +91 9740164456	
Name: BHAVYA N U USN No.: 4AI22IS008 Email id: bhavyabhavyanu08@gmail.com Mobile No: +91 8618915473									
Name: NAGAMMA V T USN No.: 4AI22IS025 Email id: nagammatimmanagoudra@gmail.com Mobile No.: +91 8073339817									
Name: SHASHANK C L USN No.: 4AI22IS041 Email id: shashank07cl@gmail.com Mobile No.: +91 8618025986									
Name: VASUDHA B M USN No.: 4AI22IS061 Email id: vasudham282@gmail.com Mobile No.: +91 9740164456									
7.	Team Leader of the Project: Name: BHAVYA N U USN No.: 4AI22IS008 Email id: bhavyabhavyanu08@gmail.com Mobile No.: +91 8618915473								
8.	Processing Fee Details (Through Online Payment only): (processing fee of Rs. 1180/-) BANK REF. NO. / UTR NO. / UPI No. (12 digits): 114267937637 TRANSACTION ID : CICAgOih-N7vRw NAME OF THE SENDER and CONTACT NUMBER: BHAVYA N U +91 8618915473 NAME OF THE BANK: CANARA BANK								

	PROCESSING FEES: Rs. 1180/- (Inclusive of 18% GST) DATE OF PAYMENT MADE: 17 th November 2025 TIME: 12:20 PM MODE OF PAYMENT MADE: UPI (GPay)																																						
9.	Date of commencement of the Project: 4 th August 2025																																						
10.	Probable date of completion of the project: 15 th February 2026																																						
11.	Timeline Structure <table border="1"> <thead> <tr> <th>Task</th><th>Start Date</th><th>End Date</th><th>Duration</th><th>Milestone achievable</th></tr> </thead> <tbody> <tr> <td>Project Initiation</td><td>4th August 2025</td><td>30th August 2025</td><td>27 Days</td><td>1. Initial Project Scope and Problem identification defined</td></tr> <tr> <td>Research Phase</td><td>1th September 2025</td><td>30th September 2025</td><td>30 days</td><td>1. Feasible study completed 2. Literature survey completed 3. Data Collection Completed</td></tr> <tr> <td>Development Phase</td><td>1st October 2025</td><td>30th October 2025</td><td>30 days</td><td>1. Detailed Design Completed 2. Modules to be identified (<i>In Progress</i>) 3. Writing the Source code for the Modules (<i>Yet to be Completed</i>)</td></tr> <tr> <td>Testing Phase</td><td>1st November 2025</td><td>30th November 2025</td><td>30 days</td><td>1. Test cases to be prepared (<i>Yet to Prepared</i>) 2. Unit test to be conducted (<i>Yet to be Conducted</i>) 3. Integration testing to be conducted (<i>Yet to be Conducted</i>)</td></tr> <tr> <td>Submission of PPT and Abstract to KSCST for conducting mid-term evaluation (Upload to Google Form)</td><td colspan="4">Middle of March – 2nd Week of April 2026</td></tr> <tr> <td>Final Review</td><td>1st December</td><td>14th January</td><td>45 Days</td><td>1. Project report to be finalised (<i>Yet to be</i></td></tr> </tbody> </table>				Task	Start Date	End Date	Duration	Milestone achievable	Project Initiation	4 th August 2025	30 th August 2025	27 Days	1. Initial Project Scope and Problem identification defined	Research Phase	1 th September 2025	30 th September 2025	30 days	1. Feasible study completed 2. Literature survey completed 3. Data Collection Completed	Development Phase	1 st October 2025	30 th October 2025	30 days	1. Detailed Design Completed 2. Modules to be identified (<i>In Progress</i>) 3. Writing the Source code for the Modules (<i>Yet to be Completed</i>)	Testing Phase	1 st November 2025	30 th November 2025	30 days	1. Test cases to be prepared (<i>Yet to Prepared</i>) 2. Unit test to be conducted (<i>Yet to be Conducted</i>) 3. Integration testing to be conducted (<i>Yet to be Conducted</i>)	Submission of PPT and Abstract to KSCST for conducting mid-term evaluation (Upload to Google Form)	Middle of March – 2 nd Week of April 2026				Final Review	1 st December	14 th January	45 Days	1. Project report to be finalised (<i>Yet to be</i>
Task	Start Date	End Date	Duration	Milestone achievable																																			
Project Initiation	4 th August 2025	30 th August 2025	27 Days	1. Initial Project Scope and Problem identification defined																																			
Research Phase	1 th September 2025	30 th September 2025	30 days	1. Feasible study completed 2. Literature survey completed 3. Data Collection Completed																																			
Development Phase	1 st October 2025	30 th October 2025	30 days	1. Detailed Design Completed 2. Modules to be identified (<i>In Progress</i>) 3. Writing the Source code for the Modules (<i>Yet to be Completed</i>)																																			
Testing Phase	1 st November 2025	30 th November 2025	30 days	1. Test cases to be prepared (<i>Yet to Prepared</i>) 2. Unit test to be conducted (<i>Yet to be Conducted</i>) 3. Integration testing to be conducted (<i>Yet to be Conducted</i>)																																			
Submission of PPT and Abstract to KSCST for conducting mid-term evaluation (Upload to Google Form)	Middle of March – 2 nd Week of April 2026																																						
Final Review	1 st December	14 th January	45 Days	1. Project report to be finalised (<i>Yet to be</i>																																			

		2025	2026		<i>Finalized)</i> 2. Final Power Point presentation to be prepared <i>(Yet to be Prepared)</i>
	Project Submission	15 th January 2026	15 th February 2026	32 days	1. Presentation of final PPT <i>(Yet to be Presented)</i> 2. Submission of final project report <i>(Yet to be Submitted)</i> 3. Demonstration of final project <i>(Yet to be Demonstrated)</i>
	Submission of Project Completion Report to KSCST (Upload to Google Form)	April / May 2026			

12.	Scope / Objectives of the project: This project focuses on developing a complete AI-based lung disease detection system that uses deep learning for analyzing chest X-ray images. The system is designed to classify 14 lung diseases, support multi-label predictions, and provide results through a Flask web interface. The project covers data preprocessing, model training, deployment, and real-time prediction. It can be used by healthcare professionals for faster diagnosis, research purposes, and as a foundation for future predictive medical applications.
13.	Methodology: In our project, the methodology explains how we built a deep learning system that can detect multiple lung diseases from chest X-ray images. It describes the steps we followed—from collecting and preprocessing the X-ray data, to designing and training our ResNet18 model, and finally integrating it into a Flask web app. This structured method helped us turn raw medical images into accurate and usable predictions through an efficient, automated system.

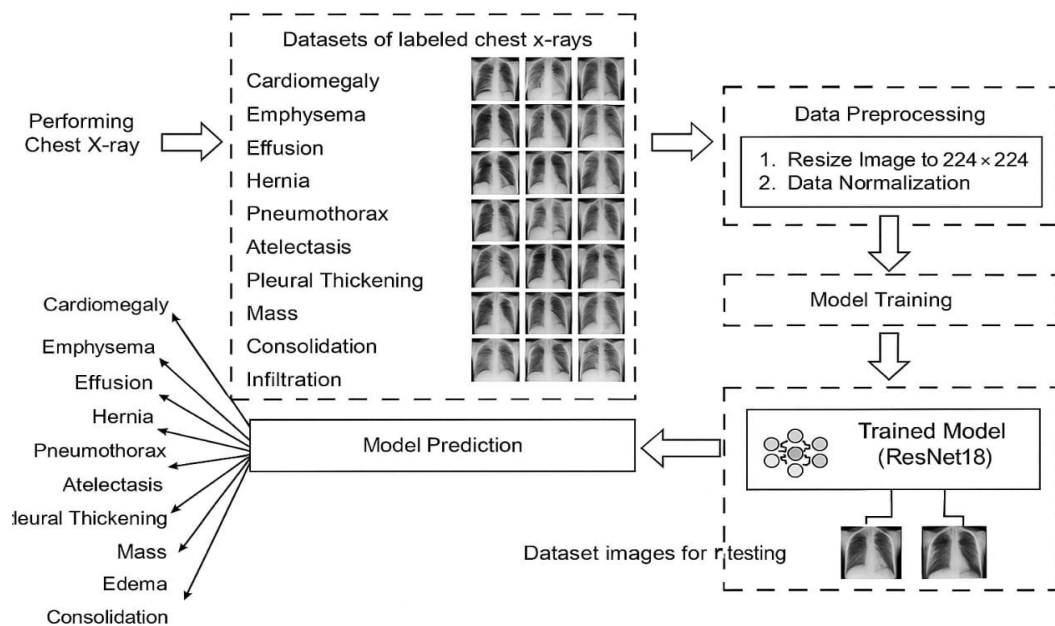


Figure 1: System Architecture

Data Collection: Chest X-ray images are collected from publicly available medical datasets such as NIH ChestX-ray14. These datasets contain thousands of X-ray images labelled with multiple lung diseases. The collected images form the training and testing base for building the deep learning model.

Data Preprocessing: All chest X-ray images undergo preprocessing to match the input format required by ResNet18. The images are resized to 224x224 pixels, normalized, and converted into tensors. This step ensures consistency, reduces noise, and prepares the images for training and prediction. The dataset is then divided into training and testing sets to evaluate performance.

Model Design and Training: A pre-trained ResNet18 model is used as the backbone architecture. The final fully connected layer is replaced with a new layer having 14 output neurons, representing the 14 lung diseases. A Sigmoid activation is used to support multi-label classification. The model is trained using Binary Cross-Entropy loss and optimizers such as Adam or SGD.

Model Evaluation: After training, the model is evaluated using the testing dataset. The performance is checked by comparing predicted results with actual labels. Metrics like accuracy, probability scores, and disease detection correctness are analysed. This step confirms how well the model generalizes to unseen X-ray images.

Model Prediction: The trained model is deployed through a Flask web application. When a user uploads a chest X-ray image, it goes through the same preprocessing steps and is then passed into the trained model. The model outputs probability values for each disease, and diseases with

	probability ≥ 0.5 are marked as detected. The webpage displays the uploaded image along with predicted diseases and their probabilities.
14.	Expected Outcome of the project: This project is designed to produce an intelligent system capable of automatically detecting multiple lung diseases from chest X-ray images using deep learning. By training a modified ResNet18 model on labeled medical images, the system is expected to generate accurate probability-based predictions for 14 different lung diseases. The final outcome also includes a fully functional Flask web application that allows users to upload an X-ray and instantly view the predicted diseases along with their confidence scores. With fast processing and clear results, the system aims to support early diagnosis, reduce the workload of medical professionals, and provide an efficient, automated tool for lung disease identification.
15.	Is the project proposed relevant to the Industry / Society or Institution? Yes / No: Yes If yes, please provide details of the industry / institution and contact details: This project is highly relevant to the industry, society, and institutions because it helps in early and accurate detection of lung diseases using deep learning. It reduces the workload of radiologists and supports faster diagnosis in hospitals and clinics. The system assists doctors in making better clinical decisions and improves patient care. It also benefits society by enabling quicker identification of serious lung conditions. For institutions, it provides a strong base for research and development in medical AI and healthcare technologies. (Note: Preference will be given to those projects relevant to the industry / institution. Hence be specific in giving detailed information). Is the industry extending support - technology / funds / use the final product, please specify.
16.	Can the product or process to be developed in the project be taken up for filing a Patent? Yes / No: No Prior Art search done? Yes/No: No Note: If your answer is “Yes”, you may contact Patent Information Centre of KSCST. For more details, email: pic@kscst.org.in

17.	Budget details (break-up details should be given): Note: KSCST will provide nominal grant support for carrying out the project by students if selected by the project selection committee. <table border="1" data-bbox="308 275 1364 607"> <thead> <tr> <th>Budget</th> <th>Amount</th> </tr> </thead> <tbody> <tr> <td>a) Materials / Consumables (Please specify)</td> <td>5000.00</td> </tr> <tr> <td>b) Labor (Describe)</td> <td>2000.00</td> </tr> <tr> <td>c) Travel (Describe)</td> <td>1000.00</td> </tr> <tr> <td>e) Miscellaneous (Please specify)</td> <td>1000.00</td> </tr> <tr> <td>Total</td> <td>9,000.00</td> </tr> </tbody> </table>	Budget	Amount	a) Materials / Consumables (Please specify)	5000.00	b) Labor (Describe)	2000.00	c) Travel (Describe)	1000.00	e) Miscellaneous (Please specify)	1000.00	Total	9,000.00
Budget	Amount												
a) Materials / Consumables (Please specify)	5000.00												
b) Labor (Describe)	2000.00												
c) Travel (Describe)	1000.00												
e) Miscellaneous (Please specify)	1000.00												
Total	9,000.00												
18.	Any other technical details (Please specify): Hardware Specification: ➤ Processor ➤ RAM ➤ Hard disk ➤ Keyboard ➤ Mouse ➤ Display Software Specification: ➤ Web Browser ➤ Python ➤ Anaconda-jupyter												
19.	SPP Coordinator (Identified by the college): Name: Dr. Kiran Kumar B M Email id: kirangowda.82@gmail.com Contact No.: +91 78293 82348												

(Dr. Kiran Kumar B M)
(Name & Signature of
SPP Coordinator with Seal)

Email id: kirangowda.82@gmail.com
Contact No.: +91 78293 82348

(Mr. Rakesh S Raj)
Name and signature of the
Project Guide with Seal:

Email id: rakeshraj.ait@gmail.com
Contact No.: +91 9902770688

DECLARATION

(From Project Students)

(To scan this page and enclose in the project proposal)

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: **“Multi Class Classification of Lung Diseases on Chest X-ray Images Using Deep learning”**, Branch: **Information Science & Engineering**, College: **Adichunchanagiri Institute of Technology**) are true and correct to the best of our knowledge and belief. We undertake to inform the Karnataka State Council for Science and Technology (KSCST) of any changes to the project title or team members' names immediately through our project guide or the SPP Coordinator of our institution.

Additionally, we declare that the project work is original and not a result of copying or purchasing. We are committed to completing the project independently, with support from our faculty and project guide, while utilizing the facilities provided by the college. We will not engage in plagiarism and pledge to be sincere and dedicated in executing and completing the project as proposed.

We understand that if any of the above information is found to be false, untrue, or misleading, we may be held liable. We authorize the sharing of the project information contained in this proposal with KSCST, Bengaluru.

We acknowledge that the project team must exhibit and demonstrate the project, participate in the mid-term evaluation of sanctioned projects, and engage with experts. Additionally, we must exhibit the project at the Annual State-Level Poster Presentation and Exhibition, if selected. Should our team fail to participate in the mid-term evaluation or the Annual Exhibition (if selected), we understand that the supported project funds will be returned to KSCST.

We also enclose the endorsement form for KSCST, Bengaluru.

Name of the students with USN No.

Signature with date

1. Bhavya N U (4AI22IS008)
2. Nagamma V T (4AI22IS025)
3. Shashank C L (4AI22IS041)
4. Vasudha B M (4AI22IS061)

(Dr. Kiran Kumar B M)

(Name & Signature of

SPP Coordinator with Seal)

Email id: kirangowda.82@gmail.com

Contact No.: +91 78293 82348

(Mr. Rakesh S Raj)

Name and signature of the

Project Guide with Seal:

Email-id: rakeshraj.ait@gmail.com

Contact No.: +91 9902770688

ENDORSEMENT

(From College, endorsement to be taken in the institution / Department Letter head)

(To scan this page and enclose in the project proposal)

This is to certify that 1) Mr. / Ms....., 2) Mr. / Ms.
3) Mr. / Ms., 4) Mr. / Ms., are bonafide student(s) of Department of, in the degree program of our institution. If the project proposal submitted by these students under the 49th series of Student Project Programme is selected by KSCST, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the mid-term evaluation of project and in the Annual State-Level Poster Presentation and Exhibition (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in mid-term evaluation of sanctioned projects or fails to attend the Annual State-Level Poster Presentation and Exhibition (if selected), the supported project amount will be returned to KSCST.

(Name & Signature of
Project Guide with Seal)

Email id:

Contact No.:

(Name & Signature of
SPP Coordinator with Seal)

Email id:

Contact No.:

(Signature of **HoD**
with Seal)

Email id:

Contact No.:

(Signature of the Principal
with Seal)

Email id:

Contact No.:

DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(**Note:** Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848, email: finance@kscst.org.in

1. TITLE OF THE PROJECT	:	"Multi Class Classification of Lung Diseases on Chest X-ray Images Using Deep learning"
2. NAME OF THE TEAM LEADER	:	BHAVYA N U
3. EMAIL ID	:	bhavyabhavyanu08@gmail.com@gmail.com
4. CONTACT MOBILE NO.	:	+91 8618915473

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	114267937637
6. TRANSACTION ID	:	CICAgOih-N7vRw
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	BHAVYA N U +91 8618915473
8. NAME OF THE BANK	:	CANARA BANK
9. PROCESSING FEES	:	Rs. 1,180/- (Inclusive of 18% GST)
10. DATE OF PAYMENT MADE	:	17 th November 2025
11. TIME	:	12:20 PM
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	UPI (GPay)

(Ms. Bhavya N U)
(Name & Signature of
the team leader)

(Mr. Rakesh S Raj)
(Name & Signature of
Project Guide / SPP Co-ordinator / HOD
with Seal)

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

49th SERIES OF STUDENT PROJECT PROGRAMME (SPP)

(Note: This page is for information about bank details of KSCST to the student team and college / institution and not to include this page in the project proposal softcopy)

BANK ACCOUNT DETAILS OF KSCST

Name and address of the Institution	Karnataka State Council for Science and Technology, IISc Campus, Bangalore -560012
Account holder's name / Designation	Secretary, Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank, IISc Campus Branch, Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank, Indian Institute of Science, Bangalore-560012

BANK DETAILS

Name of the Agency	Karnataka State Council for Science and Technology IISc Campus, Bangalore - 560012
Account holder's name / Designation	Secretary , Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank IISc Campus Branch Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank Indian Institute of Science Bangalore-560012

Note: For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848 during office hours between 3.00 pm to 5 pm, email: finance@kscst.org.in with a copy to spp@kscst.org.in



केनरा बैंक Canara Bank

भारत सरकार का उपक्रम

A Government of India Undertaking

सिंडिकेट Syndicate

KARNATAKA STATE COUNCIL FOR SCIENCE



0683201000024@cnrb

BHIM
BHARAT INTERFACE FOR MONEY

UPI
UNIFIED PAYMENTS INTERFACE



भारत 2023 INDIA

वसुधैव कुटुम्बकम्

ONE EARTH • ONE FAMILY • ONE FUTURE

Note: For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848 during office hours between 3.00 pm to 5 pm, email: finance@kscst.org.in with a copy to spp@kscst.org.in